

Lecture 23

So far: Experiment, S , $p: S \rightarrow [0,1]$, $E \subseteq S$, $P(E)$
given, calculate $P(E)$

Now: Given $P(E_1), \dots, P(E_k)$ for some events $E_1, \dots, E_k$,
what can I deduce about $P(F)$?

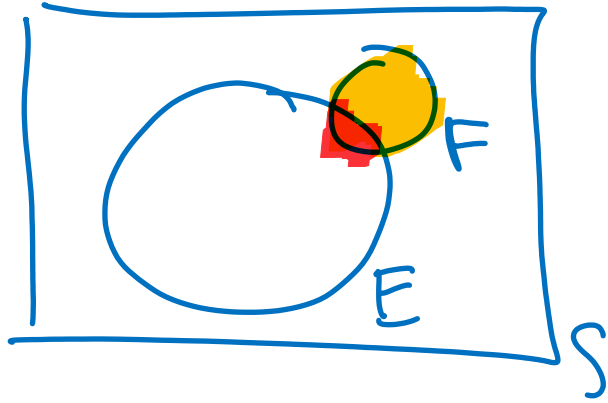
I Conditional Probability II Bayes' Rule III Independence

Defn: If E and F are events and $P(F) > 0$ then
the probability of E given F (E conditioned
on F) is defined as

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

Interpretation: "likelihood of E given that F has occurred"

Cartoon:



"Creates a new sample space which zooms in on F ."

Two Effects:

- ① Restrict the sample space to outcomes in F .
- ② Rescale $p(s)$ for $s \in E \cap F$:

$$\sum_{s \in F} \frac{p(s)}{P(F)} = \frac{P(F)}{P(F)} = 1.$$

$$P(E|F) = \sum_{s \in E \cap F} \frac{p(s)}{P(F)}$$

Special Cases:

$$\bullet P(F|F) = \frac{P(F \cap F)}{P(F)} = 1.$$

$$\bullet \text{ If } E \cap F = \emptyset \text{ then } P(E|F) = \frac{P(E \cap F)}{P(F)} = 0. \quad \text{mutually exclusive.}$$

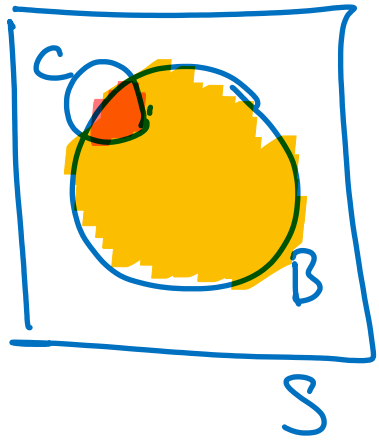
Ex¹: Expt: Pick a person in the USA (300×10^6 people) uniformly at random.

$$S = \{1, \dots, 300 \times 10^6\}.$$

$$C = \{s \in S : s \text{ is a Cal ugrad}\} \quad |C| = 30 \times 10^3$$
$$P(C) = \frac{|C|}{|S|} = 10^{-4}.$$

$$B = \{s \in S : s \text{ lives in Berkeley}\} \quad |B| = 200 \times 10^3$$
$$P(B) = \frac{|B|}{|S|} = \frac{2}{3000}.$$

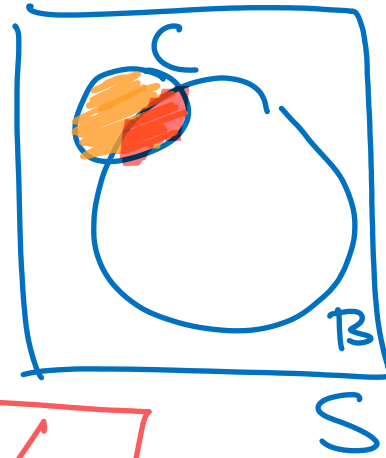
$$B \cap C = \left\{ s \in S : \begin{array}{l} s \text{ is a Cal student} \\ s \text{ lives in Berkeley} \end{array} \right\} \quad |B \cap C| = 20 \times 10^3$$
$$P(B \cap C) = \frac{2}{30,000}$$



$$P(C|B) = \frac{P(C \cap B)}{P(B)}$$

$$= \frac{2/30,000}{2/3,000} = \frac{1}{10}$$

i.e. a randomly chosen ^{"zooming in"} resident of Berkeley is a Cal student with probability 10%.



$$P(B|C) = \frac{P(B \cap C)}{P(C)}$$

$$= \frac{2/30,000}{1/10,000} = \underline{\underline{2/3}}$$

i.e. a randomly chosen Cal student is a Berkeley resident with probability 66%.

Consequences of Defn:

Product Rule:

(Rearrange formula)

$$\begin{aligned} P(E \cap F) &= P(E|F)P(F) \\ &= P(F|E)P(E) \end{aligned}$$

$P(F) > 0$

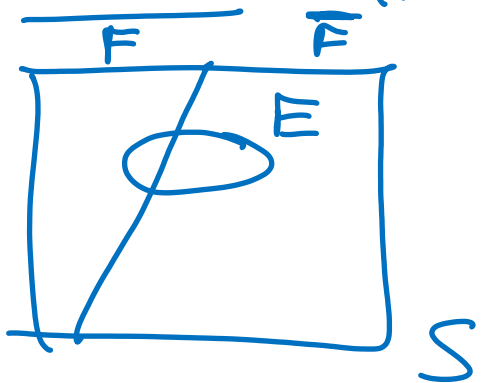
$P(E) > 0$

Remark: If $P(F) = 0$ then $P(E \cap F) = 0$.

Law of Total Probability: If E and F are events and $P(F) > 0$, $P(\bar{F}) > 0$ then

$$P(E) = P(E|F)P(F) + P(E|\bar{F})P(\bar{F})$$

Proof:



$$P(E) = P((E \cap F) \cup (E \cap \bar{F})).$$

sum
=
rule
product
=
rule

$$P(E \cap F) + P(E \cap \bar{F})$$

$$P(E|F)P(F) + P(E|\bar{F})P(\bar{F}). \quad \square$$

II Bayes' Rule

Theorem: If E, F are events with $P(E) > 0$, $P(F) > 0$
then

$$P(E|F) = P(F|E) \cdot \frac{P(E)}{P(F)}.$$

Proof: By the product rule,

$$P(E \cap F) = P(E|F)P(F) = P(F|E)P(E).$$

Rearrange.

□

Ex: COVID-19. RAT : test. $P(\bar{C}) = 0.99$.

Facts: • 1% of the campus has COVID. $P(C) = 0.01$

• If you have COVID, test reports yes 80% of the time.
 $P(T|C) = 0.8$

false positive • If you don't have COVID, it reports yes 5% of the time.
 $P(T|\bar{C}) = 0.05$

Expt: Pick a random person on campus and test them.

(Q1) What is the probability of a positive test?

(Q2) If person tests positive, what is the probability they have COVID?

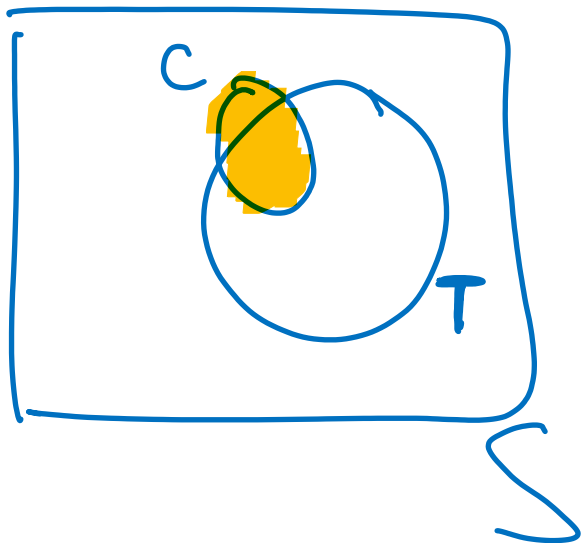
Events: $C = \{ \text{person has COVID} \}$ $\swarrow$ property of outcome
 $T = \{ \text{person tests positive} \}$

$$\begin{aligned}
 (A_1) \quad P(T) &=^{\text{total}} P(T|C)P(C) + P(T|\bar{C})P(\bar{C}) \\
 &= (0.8)(0.01) + \underline{(0.05)(0.99)} \\
 &= 0.008 + 0.0495 = \underline{0.0575}
 \end{aligned}$$

i.e. 5.75% will test positive.

$$\begin{aligned}
 (A_2) \quad P(C|T) &=^{\text{Bayes}} \frac{P(T|C)P(C)}{P(T)} \\
 &= \frac{(0.8)(0.01)}{(0.0575)} = \underline{\underline{0.139}}.
 \end{aligned}$$

i.e. if you test positive, you have COVID with prob. 13.9%.



- For a rare disease, false positives are common.
- Cannot deduce $P(C|T)$ given only $P(T|C)$, $P(T|\bar{C})$.

- Conditioning can increase or decrease $P(\cdot)$.

III Independence

Defn: Two events E and F are independent if

$$P(E \cap F) = P(E)P(F)$$

(Equivalently:

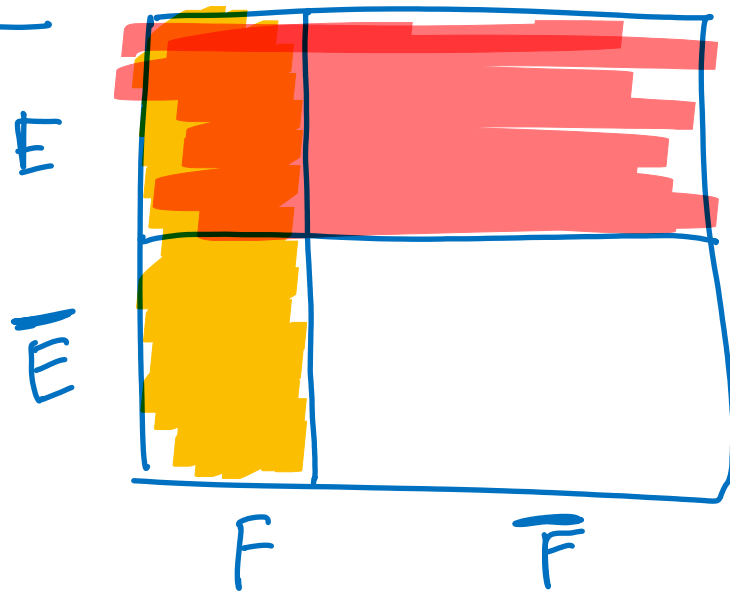
$$P(E|F) \cancel{P(F)} = P(E) \cancel{P(F)} \quad P(F) > 0$$

$$P(F|E) = P(F) \quad P(E) > 0$$

)

"Conditioning on F does not change $P(E)$!"

Cartoon:



Independent:

$$P(E) = \frac{P(E \cap F)}{P(F)}$$

Red fraction is the same
in S and in F .

ex4: Flip 2 fair coins. $S = \{H, T\}^2$

$$E_1 = \{ (s_1, s_2) : s_1 \text{ is } H \}$$

$$E_2 = \{ (s_1, s_2) : s_2 \text{ is } H \}$$

	E_1	
E_2	HH	TH
	HT	TT

$$P(E_1) = \frac{1}{2}$$

$$P(E_2) = \frac{1}{2}$$

$$P(E_1 \cap E_2) = P(\{HH\}) = \frac{1}{4} = P(E_1)P(E_2)$$

$\therefore$ They are independent.

ex5: Flip 3 independent fair coins.

hypothesis $\star$

Events involving C_1
are independent of events
involving $C_2, C_3, \dots$

$$E_1 = \{ \text{Coin 1 is H} \}, \quad E_2 = \{ \text{Coin 2 is H} \}.$$

$$E_{\text{odd}} = \{ \text{an odd \# of coins are H} \}.$$

- E_1 and E_2 are independent.

① Specified in the defn of the expt.

② Calculate: $P(E_1 \cap E_2) = 1/4 = P(E_1)P(E_2)$.

- Are E_1 and E_{odd} independent?

$$P(E_1) = 1/2.$$

$$P(E_{\text{odd}}) = P(\{\text{exactly 1 H}\}) + P(\{\text{exactly 3 H}\})$$

$$= P(\{HTT, THT, TTH\}) + P(\{HHH\})$$

$$= \frac{3}{8} + \frac{1}{8} = \underline{\underline{\frac{1}{2}}}$$

$$\begin{aligned} P(\{T H H\}) &= P(C_1=T) P(C_2=H) P(C_3=H) \\ &= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \end{aligned}$$

$$P(E_1 \cap E_{\text{odd}}) = P(\{HTT, HHH\})$$

$$= \frac{2}{8} = \frac{1}{4}$$

$$= \frac{1}{2} \cdot \frac{1}{2} = P(E_1) P(E_{\text{odd}})$$

$\therefore E_1$ and E_{odd} are independent. ← Conclusion.

Remark: Independence is a property of two events.
 it can be a conclusion (i.e. the result of a calculation)
 or it can be an assumption (part of the defn of the expt).

Independence can be obvious or hidden.

Monty Hall:

Last time: tree.

Now:

E_1 : Car is behind door 1.

M_2 : Monty picks door 2.

Want: $P(E_1 | M_2) \stackrel{\text{Bayes}}{=} P(M_2 | E_1) \cdot \frac{P(E_1)}{P(M_2)} = P(M_2 | E_1).$

Given: $P(E_1) = 1/3.$

$P(M_1) = P(M_2) = P(M_3) = 1/3$ by symmetry

< this is not going to simplify the problem.
will try again next time >